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Handy mnemonic



LEO the lion says GER

LEO - Loss of Electrons is Oxidation

GER - Gain of Electrons is Reduction

14.4 Types of Respiration

Respiration is classified into two types as aerobic and anaerobic respiration (Figure 14.4)

14.4.1 Aerobic respiration

Respiration occurring in the presence of oxygen is called **aerobic respiration**. During aerobic respiration, food materials like carbohydrates, fats and proteins are completely oxidised into CO_2, H_2O and energy is released. Aerobic respiration is a very complex process and is completed in four major steps:

1. Glycolysis
2. Pyruvate oxidation (Link reaction)
3. Krebs cycle (TCA cycle)
4. Electron Transport Chain (Terminal oxidation).

14.4.2 Anaerobic respiration

In the absence of molecular oxygen glucose is incompletely degraded into either ethyl alcohol or lactic acid (Table 14.1). It includes two steps:

1. Glycolysis
2. Fermentation

14.5 Stages of Respiration

1. Glycolysis-conversion of glucose into pyruvic acid in cytoplasm of cell.
2. Link reaction-conversion of pyruvic acid into acetyl coenzyme-A in mitochondrial matrix.
3. Krebs cycle-conversion of acetyl coenzyme A into carbon dioxide and water in the mitochondrial matrix.

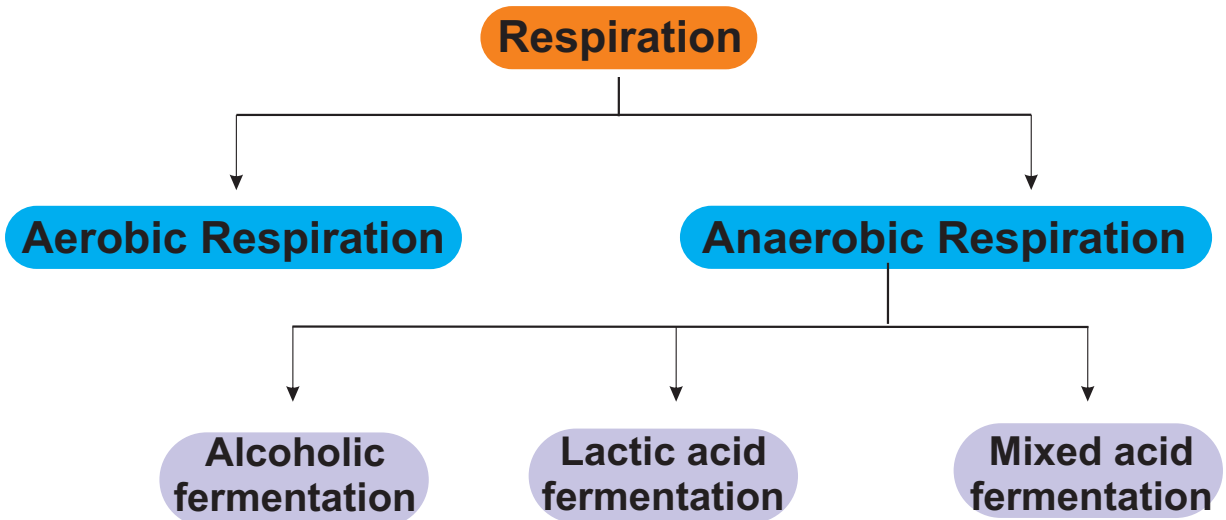


Figure 14.4: Types of Respiration